Public Access and Dissemination Plan

OligoTox-Hydrocephalus dataset · NIH/NCATS Oligonucleotide Toxicity Open Data Challenge, Phase 2

This plan describes how the OligoTox-Hydrocephalus dataset is licensed, made publicly accessible and disseminated, and — as the Challenge requires — how the U.S. Government can allow interested parties to use it even if the submitting team does not itself continue to.

1 What is covered

Artifact	Description
data/oligos.csv	50 oligonucleotides — identity and design predictors
data/measurements.csv	1,324 graded hydrocephalus / CSF-dynamics records
data/modifications.csv	202 per-position chemistry records
data/sources.csv	189 source provenance registry
data/trial_registry.csv	155 trials with posted results
OligoTox-Hydrocephalus_Dataset.xlsx	the whole dataset as one workbook, 9 sheets
SCHEMA.md, scripts/data_dictionary.py	schema and data dictionary
scripts/, qc/, ml/	build, QC and analysis code
sources/raw/	every retrieved source payload

The dataset is a curation of already-published data. It contains **no human-subjects data, no personally identifiable information and no protected health information** — only aggregate experimental and clinical values and design descriptors drawn from public literature, regulatory labels, public registries and government databases. There are therefore no privacy, consent or data-use-agreement constraints on redistribution.

2 Licensing scheme

The curated tables, schema, documentation and code are released under the **Creative Commons Attribution 4.0 International licence (CC BY 4.0)** — permissive and **irrevocable** — letting anyone, including the U.S. Government and any third party, access, reuse, redistribute, modify and build upon the dataset in perpetuity subject only to attribution. The grant is made in the LICENSE file in the dataset directory, not merely intended. A more permissive dedication such as CC0 1.0 can be substituted at NCATS's preference.

Underlying third-party full texts are never redistributed; they are referenced by DOI, PMID, NCT identifier or set id. Rights are tracked **per row** in the redistribution column, and every value it takes is used by real rows:

Value	Rows	Meaning
public_domain	1,290	US Government works — ClinicalTrials.gov, FAERS, DailyMed, eCFR. Values may be reproduced.
cc_by	8	Open-access articles; reproduce with attribution.
cc_by_nc	3	Open access, non-commercial terms.
summary_stat_only	15	Licence carries a no-derivatives term; only abstract-level summary statistics are carried.
verify	8	Reuse terms were not established in curation; a redistributor should resolve the licence before republishing those values.

No patents, trade secrets or restrictive intellectual property are or will be claimed over the dataset. There is no proprietary component whose withdrawal could remove public access. Because the licence is irrevocable once granted, third-party use does not depend on the team's continued participation.

3 Public access — hosting and persistence

- **Primary distribution:** a public code-hosting repository (github.com/naviero1/Oligos, toxicity/hydrocephalus/), openly accessible without registration.
- **Independent archival persistence and a citable identifier:** a versioned snapshot of each release will be deposited in a public long-term archive (Zenodo or an NIH-designated repository) and assigned a DOI, so the dataset remains findable and retrievable independently of the team or any single hosting account.
- **Non-proprietary formats:** UTF-8 CSV and Markdown, plus one XLSX convenience copy. No software licence is required to read any of it, and the build and QC code depends only on the Python standard library except for the workbook export and the analysis.
- **Reproducibility:** every source payload is committed, so a third party can rebuild the dataset from a clean checkout with no network access and confirm all 47 QC checks pass.

4 Dissemination and FAIR alignment

Principle	How the dataset meets it
Findable	Stable primary keys; archival DOI; descriptive metadata; a provenance registry naming every source.
Accessible	Open repository and archive; open formats; no gatekeeping, registration or request process.
Interoperable	Controlled vocabularies enforced by the QC suite; a normalised four-table schema that is deliberately column-compatible with our CNS release so endpoints can be pooled.
Reusable	CC BY 4.0, plus per-row provenance (source_id, source_ref, exact source_location) so any single value is re-verifiable, and per-row rights so lawful reuse is auditable at record level.

Documentation for reuse comprises the narrative, this plan, the methodology document, README, SCHEMA and the in-code data dictionary. Optional supporting material includes the full build and QC pipeline and the analysis code that produces the narrative's predictive-model section.

5 Continuity and U.S. Government use contingency

The Challenge requires a plan for how the U.S. Government can allow interested parties to use the solution if the team fails to use it and does not permit others to under reasonable terms. Three scenarios cover this.

- **Scenario A — normal operation.** The team maintains the repository and archival deposit and disseminates updates. Third parties use the dataset under CC BY 4.0.
- **Scenario B — the team ceases activity.** No action by the team is needed: the dataset has **already** been released under an irrevocable CC BY 4.0 licence and a copy is deposited in an independent public archive with a DOI. The U.S. Government and any interested party may continue to access, copy, host and build upon it without further permission. The permissive licence and the independent archival copy are the mechanisms that accomplish this.
- **Scenario C — explicit government grant.** The team additionally grants the U.S. Government a non-exclusive, irrevocable, royalty-free right to use, reproduce, distribute, publicly display, host copies of and **authorise others to use** the dataset and its documentation. This right survives any cessation of the team's activity.

In no scenario does third-party or Government use depend on the team's ongoing involvement, on a proprietary service, or on any revocable permission.

6 Maintenance, versioning and stewardship

Releases are versioned with a changelog and each is archived under its own DOI, so prior versions remain citable. All severity grades currently ship grade_status = provisional; a post-review release will clear that flag and record the reviewer's disposition. Corrections and additions are tracked through the repository's public history, and the QC suite runs on every revision so a release cannot silently regress. The maintainer is the submitting team via the public repository; the fallback steward is the independent public archive, which preserves the deposited snapshot and DOI irrespective of team status.

7 Compliance summary

- Consistent with NIH Scientific Data Sharing / Public Access policy and with FAIR.
- Open, irrevocable licence granted in-repository, ensuring perpetual public usability.
- No PII, PHI or human-subjects data, so no privacy or consent barrier to sharing.
- Provenance and redistribution rights tracked per row, so lawful reuse is verifiable at the record level.